



education

phd | machine learning

uc berkeley | '17-'22

research: interpretable ml

advisor: bin yu

bs | cs & math

university of virginia | '14-'17

double major

skills

llms | deep learning

data science | data cleaning

huggingface | pytorch

rule-based models | causal inference

awards

berkeley grad slam semifinalist '19, '22

pdsoros fellowship finalist '19

outstanding teaching award '18

uva rader research award '17

uva undergrad symposium winner '17

raven honor society '16-'17

icpc regional qualification '14-'16

1st place microsoft code jam '16

3rd place google games uva '17

2nd place apt puzzle competition '17

rodman scholarship '14-'17

teaching

berkeley | fall 2019

artificial intelligence: cs 188

berkeley | summer 2018

machine learning: cs 189/289

lectures to class of 80+ students

service

volunteering

basis education volunteering '19-'22

bair undergrad mentoring '18-'22

computer literacy volunteering '15-'17

organizer

iclr trustml workshop '26

area chair

ml4h '25 | xxai workshop, ml4h '24

reviewer

iclr, icml, neurips '26

arr, iclr, icml, iccv, neurips, pnas '25

iclr, icml, neurips '24

neurips '23 | acl '22

iclr, cvpr, aaai, neurips '21 | neurips '20

experience

voyage ai (acquired by mongodb)

staff research scientist | summer '26 - present

- training and interpreting large embedding models

microsoft research

senior research scientist (deep learning group) | summer '22 - summer '26

- improving the interpretability of llms
- researching knowledge discovery with llms
- building next-generation foundation models

health tech

paige ai | research scientist | summer '21 - summer '22

- interpretable deep learning in digital pathology (especially bladder cancer)

response4life | volunteer data scientist | spring '20

- helped develop, integrate, and deploy models to forecast covid-19 severity

pacmed ai | healthcare ml intern | summer '19

- developed interpretable, tabular ml models for healthcare

phd

berkeley | interpretable ml research (bin yu group) | fall '17 - spring '22

- developed post-hoc interpretation methods for neural networks
- developed interpretable models in medicine, biology, and computer vision

aws | ml fairness intern (pietro perona group) | summer '20

- testing for bias with causal matching using GANs

meta ai | computer vision intern | summer '17

- investigated unsupervised deep learning for segmentation of satellite imagery

undergrad

hhmi | computer vision intern (srini turaga group) | summer '14, '15, '16

uva | graphical models research (yanjun qi group) | fall '16 - spring '17

uva | comp. neuroscience research (william levy group) | fall '14 - fall '16

selected publications

interpretability $\times$ llms $\rightarrow$ science

- generative causal testing with llms: antonello*, cs*, et al. *nature neuroscience*, '26
- augmenting interpretable models with llms: cs*, et al. *nature comms*, '23
- interpretable embeddings by asking llms questions benara*, cs*, et al. *neurips*, '24

interpretability $\times$ rules $\rightarrow$ healthcare

- human-ai co-design for clinical models: feng*, kothari*, ..., cs. *npj digital med* '26
- fast interpretable greedy-tree sums: tan*, cs*, nasseri*, agarwal* et al. *pnas* '22
- imodels: an interpretability package: cs*, nasseri*, tan, tang, & yu. *joss* '21

interpretability $\times$ deep learning $\rightarrow$ general domain

- post-hoc attention steering for llms: zhang, cs, et al. *iclr* '24
- adaptive wavelet distillation from dnns: ha, cs, et al. *neurips* '21
- hierarchical interpretations for dnn predictions: cs*, murdoch*, & yu. *iclr* '19